

belongs to which order of the spectrum.

5 (a) $F = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2}$

$$F = \frac{(1.6 \times 10^{-19})^2}{4\pi \times 8.85 \times 10^{-12} \times (2.0 \times 10^{-15})^2}$$

$$F = 58 \text{ N}$$

- (b) It represents the work done by an external agent in bringing the two particles from a distance of $2.0 \times 10^{-15} \text{ m}$ apart to an infinite distance apart (negative because the two charges repel and hence positive work is actually done by the electric force). It is also the negative of the electric potential energy stored in this system of charges when they are $2.0 \times 10^{-15} \text{ m}$ apart.

**The limits of the distance moved were stated wrongly as from the point $2.0 \times 10^{-15} \text{ m}$ away from P_2 to P_2 .*

- (c) When the particles fuse, the products have less mass than the reactants. The mass difference becomes the kinetic energy of the products and radiation energy of the gamma rays emitted.

- 6 (a) Magnetic flux linkage of the coil = number of turns $\times$ magnetic flux density $\times$ area of the coil perpendicular to the magnetic field.